SO(3) Equivariant Neural Networks

Lecture 5 Solutions

Build a Physical SO(3) ENN

Solutions to checkpoints, discussions, and notebook exercises

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1 Solutions to lecture questions

Question 5.1

For two unit vectors, the invariant shape is their relative angle, equivalently $p \cdot E$. The dataset samples this quantity across its full range. What is restricted is only the global orientation of the pair. Therefore training covers all intrinsic two-vector shapes while withholding most rotated copies.

Question 5.2

MSE measures agreement with labeled targets, while equivariance error measures agreement with a transformation law. A model can be exactly equivariant but inaccurate, or accurate on a narrow test set without being equivariant. Reporting both distinguishes physics-structure correctness from predictive quality.

Question 5.3

p and E are $\ell = 1$ vectors. $U = -p \cdot E$ is an $\ell = 0$ scalar, produced by the $1 \otimes 1 \to 0$ path. $\tau = p \times E$ is an $\ell = 1$ vector under proper rotations, produced by the $1 \otimes 1 \to 1$ path.

Question 5.4

Relative displacements remove global translation. Spherical harmonics encode edge directions as SO(3) irreps. CG products combine those angular types equivariantly. Neighbor aggregation removes dependence on arbitrary neighbor ordering and provides permutation symmetry. These operations address different group actions and are complementary.

Question 5.5

If $E(Rr) = E(r)$ and R is orthogonal, differentiating with respect to rotated coordinates gives $\nabla_{Rr} E(Rr) = R \nabla_r E(r)$. Hence $F = -\nabla E$ transforms as $F(Rr) = RF(r)$. This assumes smooth differentiation, consistent transformation of every coordinate, and no unrotated external symmetry-breaking variable.

Question 5.6

A penalty adds an energetic cost for leaving a desired angular-momentum sector but may permit residual mixing at finite penalty strength or imperfect optimization. An architectural construction restricts the variational family itself to the chosen transformation law. The former is soft guidance; the latter is a hard function-space constraint.

Question 5.7

The accuracy half: rotated copies fill in the orientations the training band omitted, so the model no longer has to extrapolate into regions of input space it never saw. It pays for this with a worse fit on the original examples, because the same fixed capacity now has to cover a far larger input distribution.
The symmetry half: augmentation constrains the model only at the sampled points. Between them nothing enforces $f(Ru, Rv) = Rf(u, v)$, and the residual measures exactly that unconstrained behavior. Adding more copies shrinks the accuracy gap and leaves the residual at order unity,

because no finite sample can impose an identity that must hold for every element of a continuous group.

Question 5.8

Two consistent explanations: (i) the path is genuinely uninformative for this target; or (ii) the path is informative but *redundant*, with the same function reachable through the same-ℓ linear path and the other couplings, so the surviving weights already implement it.
An experiment that separates them: retrain from scratch with that path removed from the architecture. Under (i) the retrained model matches the full model and its other weights are essentially unchanged. Under (ii) it also matches, but the surviving weights move substantially to take over the job, so compare fitted weights and not only error. A sharper version removes several paths at once: redundancy predicts that error grows only once every route to the output is cut, while irrelevance predicts no growth at all. The general lesson is that zeroing one component of a trained model measures replaceability, not importance.

2 Capstone reference design

For an atomistic energy-and-force model, one reasonable sketch is:

1. Inputs: species labels ($\ell = 0$) and positions.
2. Geometry: relative vectors r_{ij} to ensure translation invariance.
3. Edge features: radial basis functions of $\|r_{ij}\|$ and $Y_{\ell m}(\hat{r}_{ij})$ for angular structure.
4. Hidden state: several channels for $\ell = 0, 1, 2$ or higher.
5. Interaction: CG tensor products followed by learned channel mixing and scalar gates.
6. Aggregation: sum neighbor messages to obtain permutation invariance.
7. Readout: atomic $\ell = 0$ energies summed to total energy.
8. Forces: $F_i = -\nabla_{r_i} E$, computed by automatic differentiation.
9. Tests: translations, rotations, permutations, energy conservation, and held-out physical configurations.

3 Notebook reference interpretation

The expected qualitative result is:

- the CG network has train and rotated-test MSE of similar scale;
- its equivariance residual stays near 10^{-15} before and after training;
- the Cartesian MLP may fit the narrow training band but degrade on broad rotations.

This demonstrates orientation generalization, not universal superiority. A fair scientific evaluation must also test unseen invariant shapes, noise, model capacity, and computational cost.